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NAAC GRADE A+ ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

DEEP LEARNING

Assignment No.: 3

Title: COMPARISON OF DEEP LEARNING ARCHITECTURES

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Project 3: Comparison of Deep Learning Architectures

Introduction

Deep Learning is a branch of Artificial Intelligence that uses neural networks with multiple layers to learn patterns from data. Different deep learning architectures are designed for different types of data and applications. This project focuses on three important architectures: Feedforward Neural Networks (FNNs), Convolutional Neural Networks (CNNs), and Recurrent Neural Networks (RNNs). The purpose of this project is to understand their basic architectures, suitable applications, and key differences.

Task 1: Basic Architecture of Deep Learning Networks

1. Feedforward Neural Network (FNN)

A Feedforward Neural Network is a basic type of neural network in which information moves in one direction, from the input layer through one or more hidden layers to the output layer. It does not have feedback or recurrent connections. The input layer receives the data, the hidden layers process the information, and the output layer produces the final result. Feedforward Neural Networks are commonly used for classification, regression, prediction, and structured or tabular data.

Input Layer → Hidden Layer(s) → Output Layer

Figure 1: Basic Architecture of a Feedforward Neural Network

2. Convolutional Neural Network (CNN)

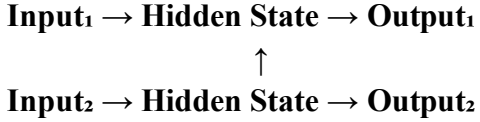
A Convolutional Neural Network is a deep learning architecture mainly designed for processing image and spatial data. CNNs use convolutional layers to detect important features such as edges, shapes, and patterns from the input. A typical CNN contains an input layer, convolutional layers, pooling layers, fully connected layers, and an output layer. CNNs are commonly used for image classification, object detection, facial recognition, and medical image analysis.

Input Image → Convolution → Pooling → Fully Connected → Output

Figure 2: Basic Architecture of a Convolutional Neural Network

3. Recurrent Neural Network (RNN)

A Recurrent Neural Network is a deep learning architecture designed to work with sequential or time-dependent data. Unlike a basic feedforward network, an RNN uses recurrent connections that allow information from previous time steps to influence later inputs. Its main components include the input, hidden state, and output. RNNs are commonly used for text processing, speech recognition, language-related tasks, and time-series prediction.



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Input₃ → Hidden State → Output₃

Figure 3: Basic Architecture of a Recurrent Neural Network

Task 2: Suitable Applications for Each Architecture

Architecture	Suitable Applications	Type of Data
Feedforward Neural Network	Classification, regression, prediction	Structured / tabular data
Convolutional Neural Network	Image classification, object detection, facial recognition, medical image analysis	Image / spatial data
Recurrent Neural Network	Text processing, speech recognition, language tasks, time-series prediction	Sequential / time-dependent data

Task 3: Comparison of Deep Learning Architectures

Feature	FNN	CNN	RNN
Structure	Input → Hidden → Output	Convolution → Pooling → Fully Connected → Output	Input → Hidden State → Output with recurrent connections
Input Data	Structured / tabular data	Image / spatial data	Sequential / time-dependent data
Main Feature	General pattern learning	Spatial feature extraction	Sequential dependency
Previous Input Memory	No	No inherent sequence memory	Yes
Common Applications	Classification, regression, prediction	Image classification, object detection	Text, speech, time series
Main Advantage	Simple and flexible	Effective for spatial patterns	Handles sequential relationships

Key Differences

- FNN processes information in a forward direction and is suitable for general prediction and classification tasks.
- CNN is designed to identify spatial patterns and is mainly used for image-related applications.
- RNN processes sequential information and uses previous information when processing later inputs.
- FNNs generally work well with structured data, CNNs with image or spatial data, and RNNs with sequential data.

Conclusion

This project provided an understanding of three important deep learning architectures: Feedforward Neural Networks, Convolutional Neural Networks, and Recurrent Neural Networks. The architectures differ in their structure, type of input data, and suitable applications. Feedforward Networks are useful for general prediction and classification, CNNs are well suited for image and spatial data, and RNNs

are suitable for sequential and time-dependent data. Understanding these differences helps in selecting an appropriate deep learning architecture for a particular problem.

Learning Outcomes

Through this project, I learned:

- The basic architecture of Feedforward Neural Networks.
- The basic architecture of Convolutional Neural Networks.
- The basic architecture of Recurrent Neural Networks.
- The suitable applications of different deep learning architectures.
- The differences between FNNs, CNNs, and RNNs.
- How the type of input data influences the choice of deep learning architecture.
- How different deep learning architectures are applied to real-world problems.